

Internship Final Report

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Major: Marine Engineering

Internship Duration: April 10th, 2025 - May 3rd, 2025

Company: Hack Secure

Domain: Cyber Security

Mentor: Mr.Nishant Prajapati

Assistant Mentor: Mr. Aman Pandey

Coordinator: Mr. Shivam Kapoor

Objectives

My primary objectives for this internship were to:

1. Develop a deep understanding of cybersecurity principles and practices.
2. Gain hands-on experience in identifying, analyzing, and mitigating security threats.
3. Enhance my skills in using cybersecurity tools and techniques in real-world scenarios.

Tasks and Responsibilities

During my internship, I was involved in the following key tasks:

- **Vulnerability Assessment:** Conducted a thorough scan of a target website to identify open ports and potential vulnerabilities.
- **Penetration Testing:** Performed brute-force attacks and directory enumeration on the website to uncover hidden directories and files.
- **Traffic Analysis:** Intercepted network traffic using Wireshark during a simulated login attempt, successfully capturing and analyzing transmitted credentials.

Intermediate Task

Task

- **Outcomes** – Summary of findings, tools used, and key results.
- **Conclusion** – Skills gained, challenges faced, and improvements.
- **Screenshot** – Proof of task completion with clear visuals.

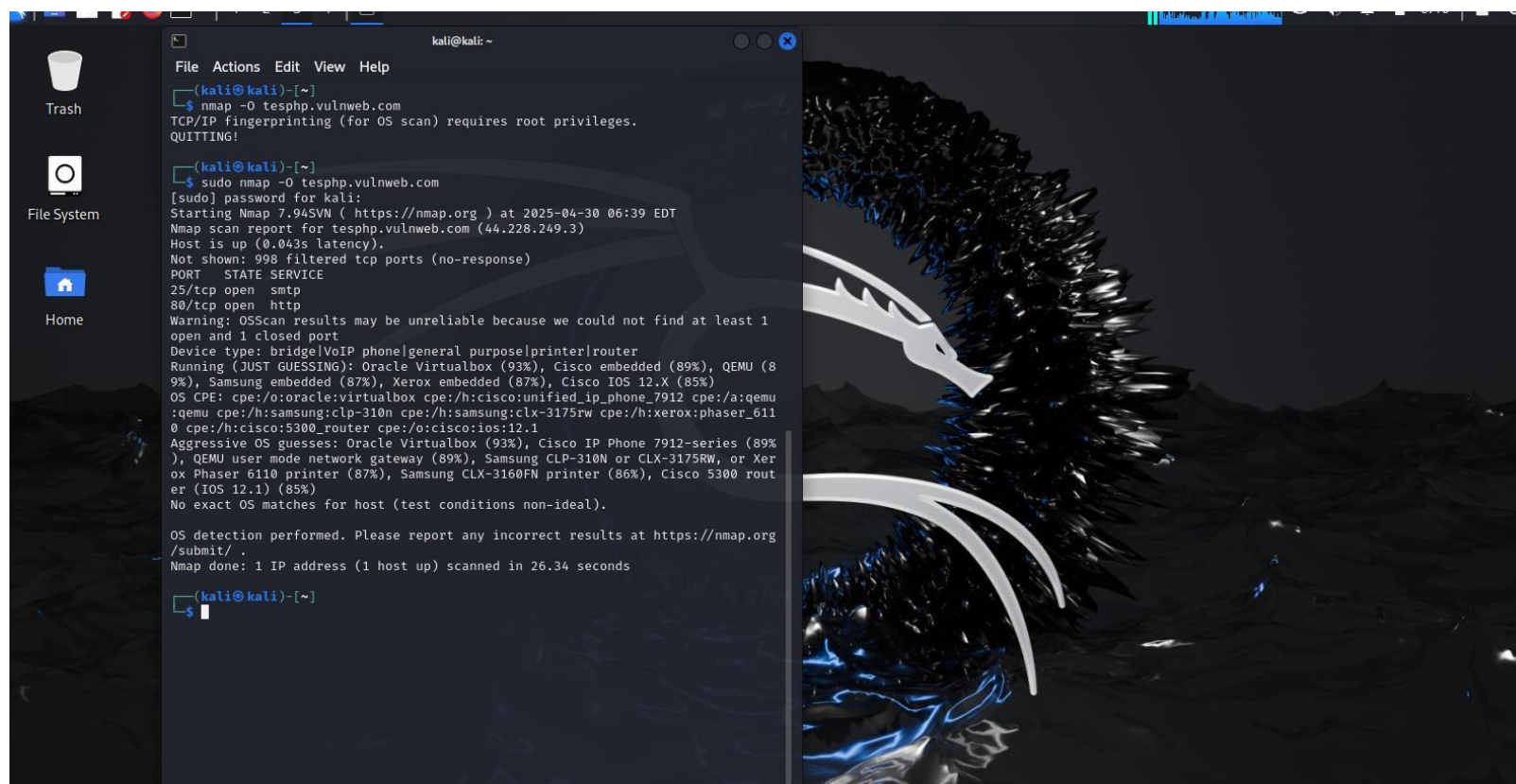
Upon receiving the url (<https://www.testphp.vulnweb.com>)

NB: the following scans and test were done using an OS (operating system) called a Kali Linux

- **Finding the open ports on the domain**

This can be done using a pre-installed tool in Kali linux [Nmap]

Command: Nmap <ipadress of the given domain> | **Nmap <ip>** NB. This is a basic Nmap scan The result should look like this.

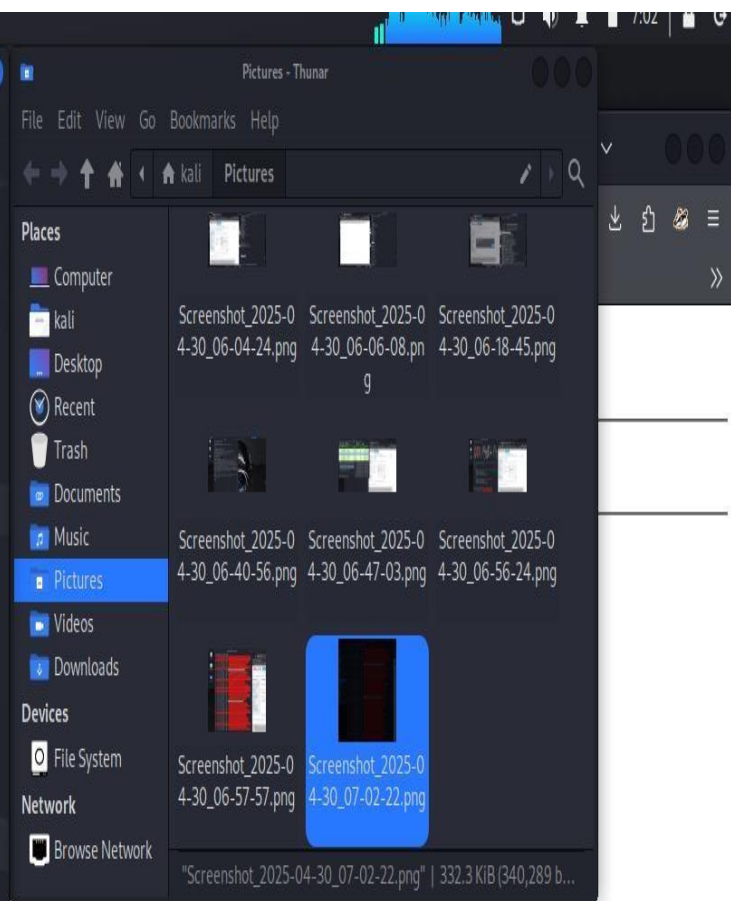
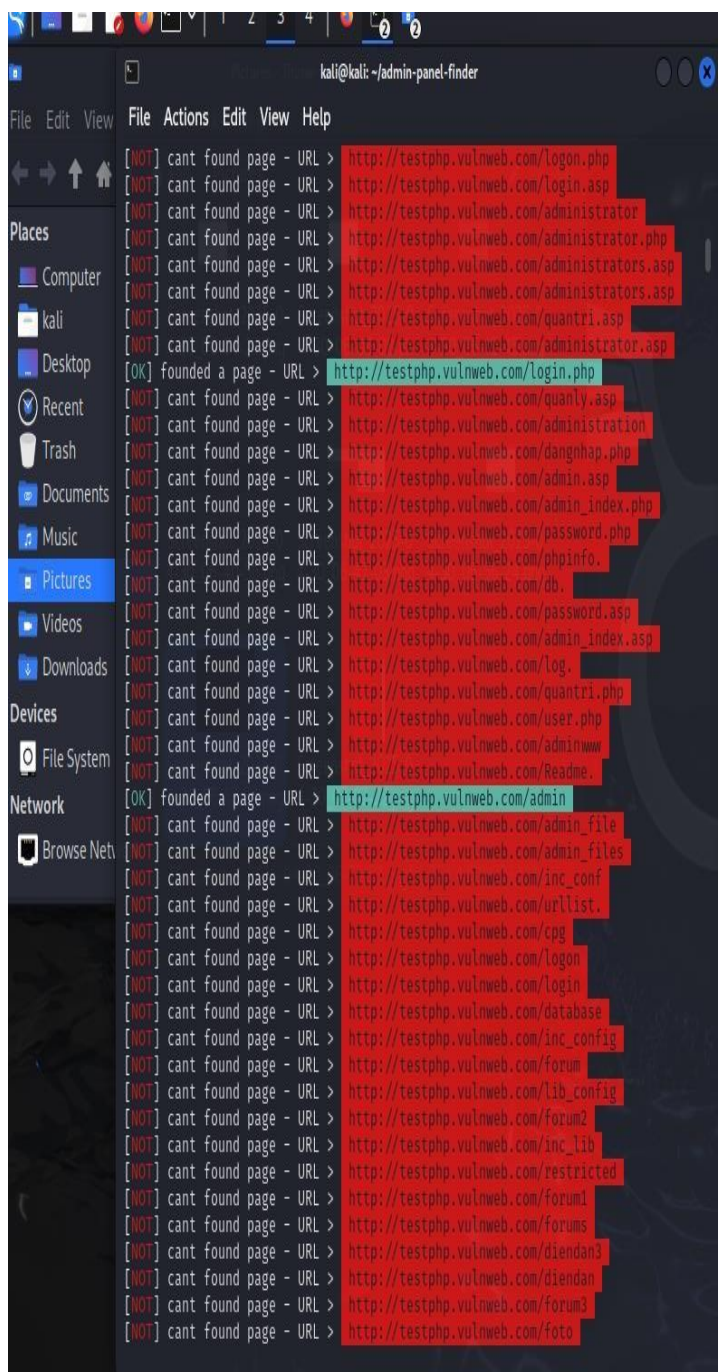


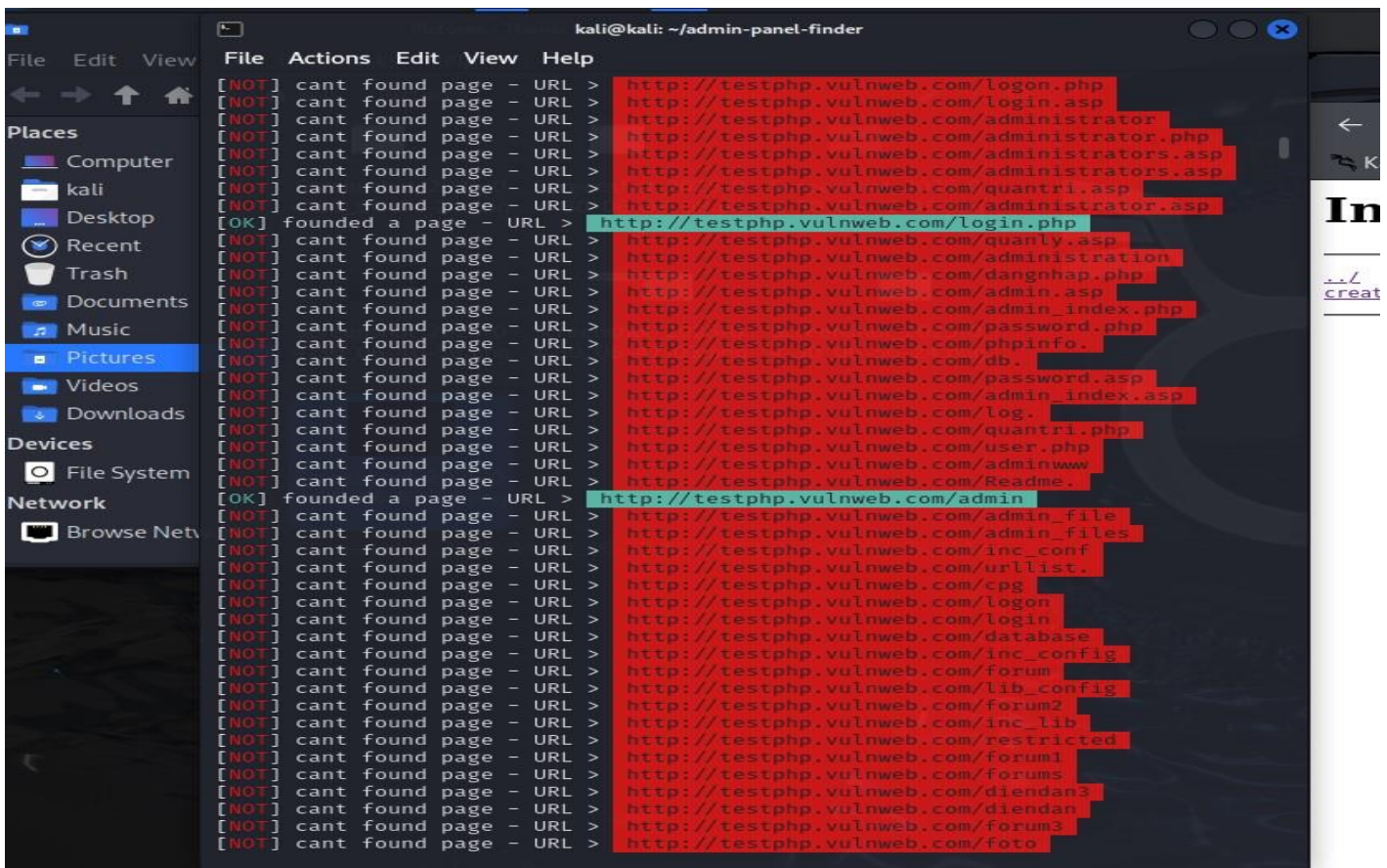
- Brute force the website <https://testphp.vulnweb.com> for sub-directories

Using the tool admin-panel-finder found in github

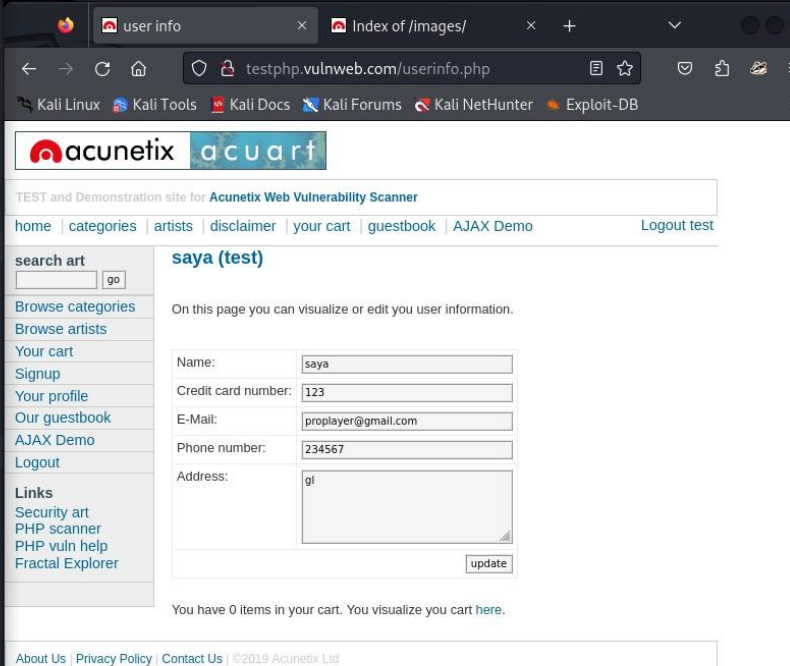
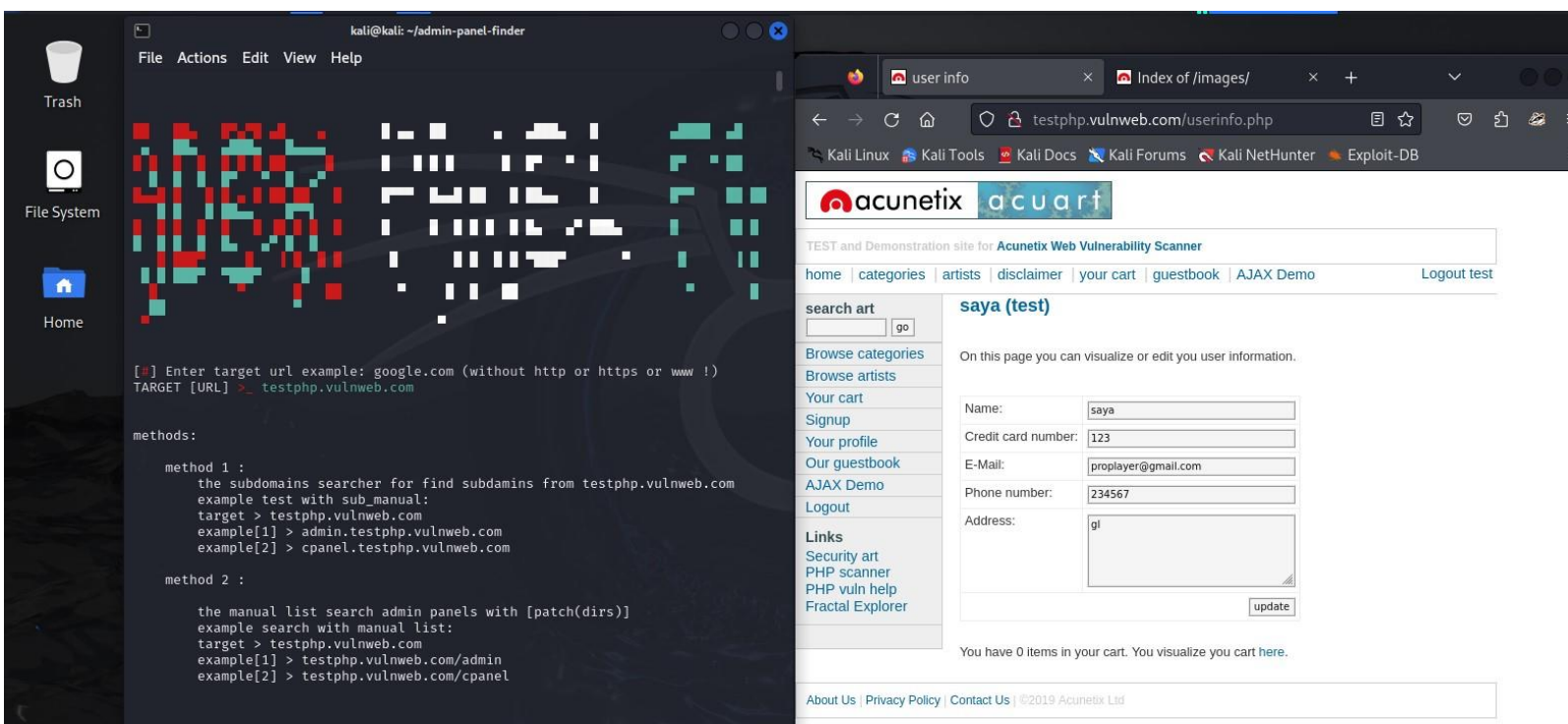
Command: python3 admin-finder.py

This was the result





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- Intercepting Network Traffic And Capturing Credentials

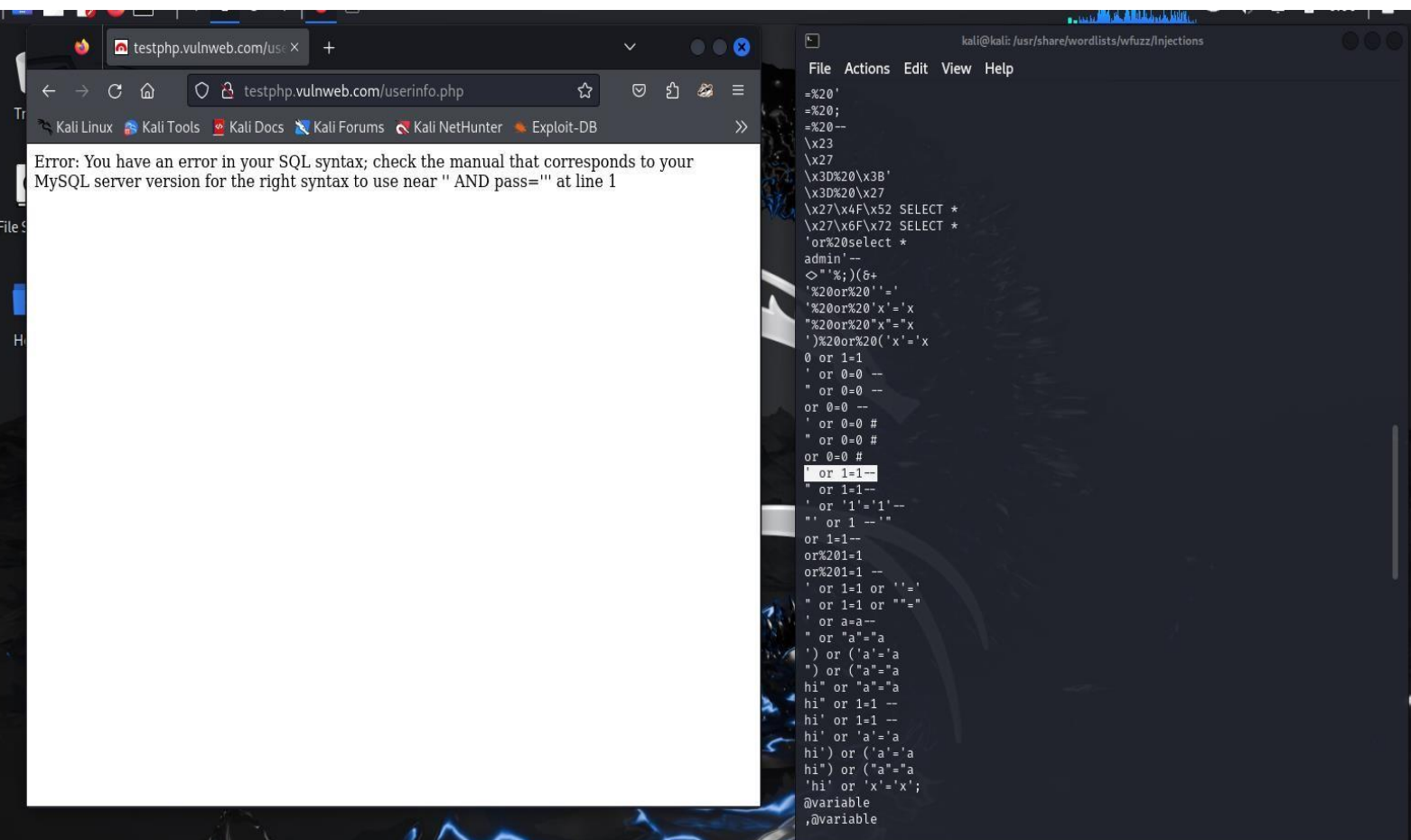
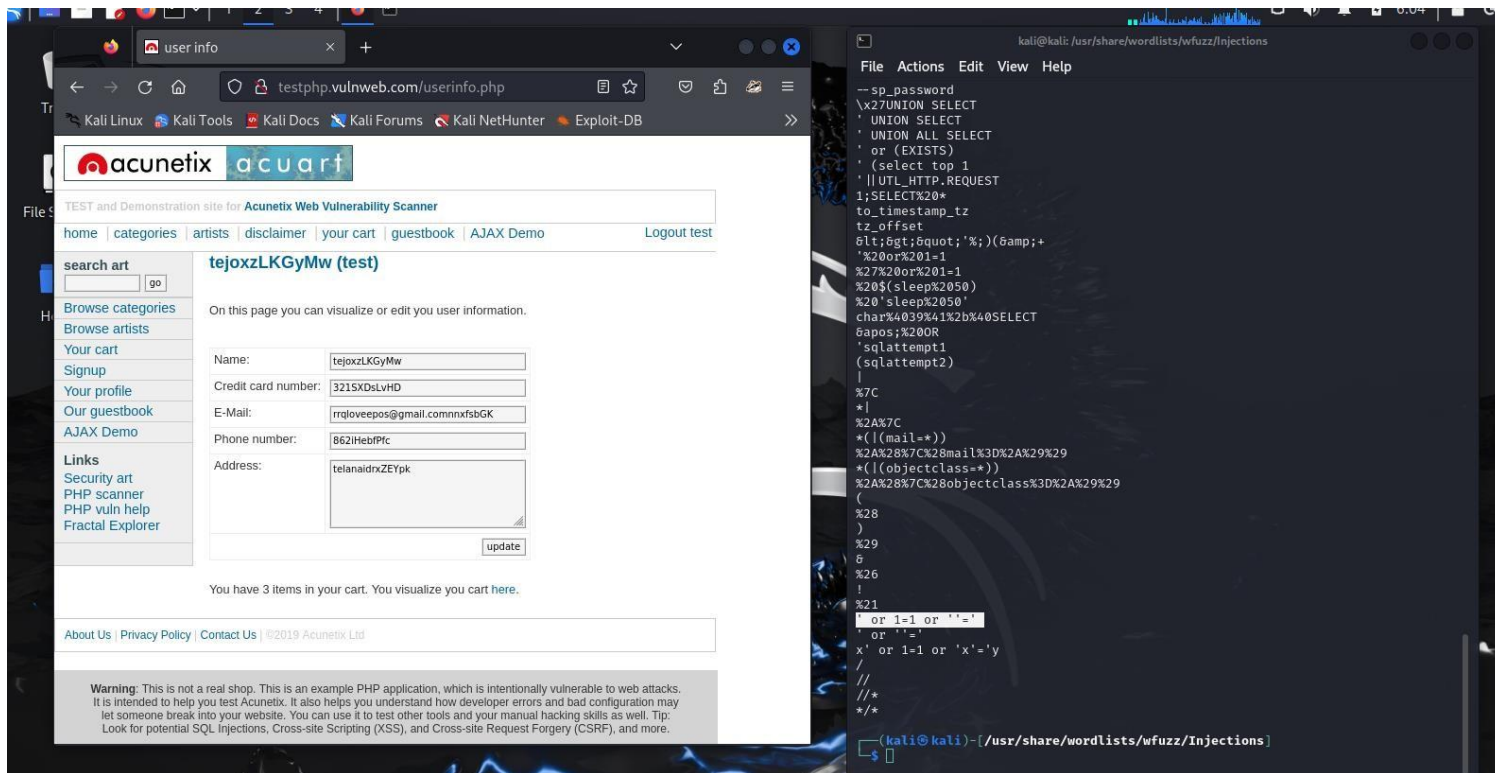
Using Wireshark with the filter: `ip.addr == <44.228.249.3>` and the result were as follows

The image shows two overlapping windows. The background window is Wireshark, displaying a packet capture on interface `eth0` with the filter `ip.addr == 44.228.249.3`. The packet list shows several TCP and HTTP packets. The selected packet (No. 14) is an HTTP POST request to `/userinfo.php` with a body containing `uname=test` and `pass=test`. The foreground window is a web browser showing the page `testphp.vulnweb.com/userinfo.php`. The page displays a user profile for 'saya (test)' with fields for Name, Credit card number (123), E-Mail (proplayer@gmail.com), Phone number (234567), and Address (gl). The page also includes a warning at the bottom: 'Warning: This is not a real shop. This is an example PHP application, which is intentionally vulnerable to web attacks.'

- Performing SQL Injections on the login page
<https://testphp.vulnweb.com/login>

The injection was done using the sql syntax `< " or 1=1 or " ' = ' >`

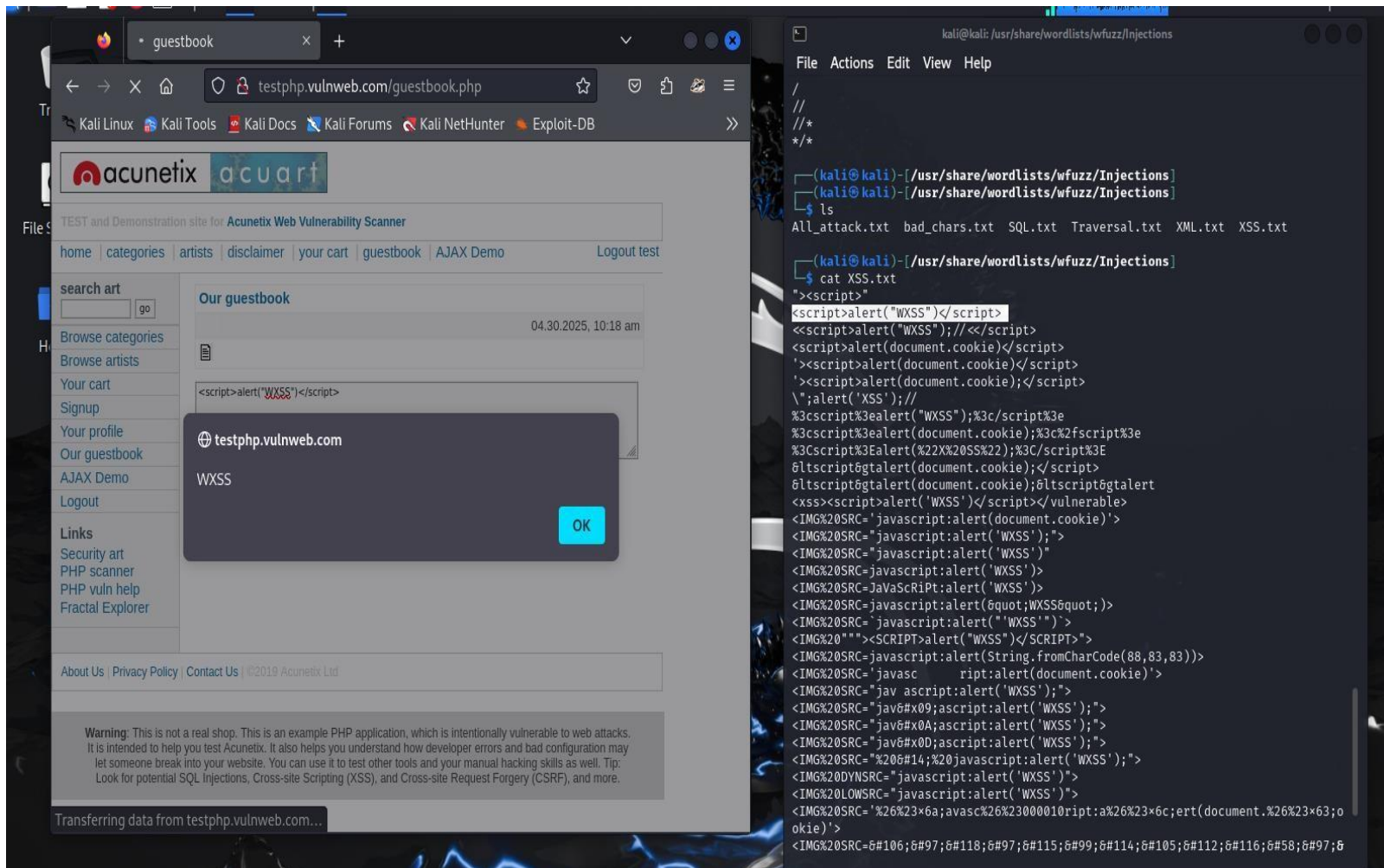
And the result of the injection was



- Injecting malicious JavaScript payloads(XSS) on the given url <https://testphp.vulnweb.com>

The Injection was done using the XSS payload (`<script>alert("WXSS")</script>`) And

the result was



Ethical Hacking Project (EH)

- Make a port scanner from the provided ip address using python


```

import socket import
subprocess import
platform
def
is_ip_alive(ip_address):
    """
    Checks if an IP address is alive using ping.

    Args:
        ip_address (str): The IP address to check.

    Returns:
        bool: True if the IP is alive, False otherwise.
    """
try:
    # Determine the operating system to use the correct ping command.
    system = platform.system()
    if system == "Windows":
        # -n 1 sends one echo request
        ping_command = ["ping", "-n", "1", ip_address]
    elif system == "Darwin" or system == "Linux":
        # -c 1 sends one echo request
        ping_command = ["ping", "-c", "1", ip_address]
    else:
        print(f"Unsupported operating system: {system}")
    return False

    # Execute the ping command and capture the output.
    process = subprocess.Popen(ping_command, stdout=subprocess.PIPE,
                                stderr=subprocess.PIPE)
    stdout, stderr = process.communicate()

    # Check the return code to determine if the ping was successful.
    if process.returncode == 0:
        return True
else:
    return False

```

```

except Exception as e:
    print(f"Error checking IP: {e}")
return False
def basic_port_check(ip_address, port):
    """
    Performs a basic TCP port check.

    Args:
        ip_address (str): The IP address to check.
        port (int): The port number to check.

    Returns:
        bool: True if the port is open, False otherwise.
    """
    try:
        with socket.socket(socket.AF_INET, socket.SOCK_STREAM)
as sock:
        sock.settimeout(1)  # Set a timeout for the connection
        attempt.
        result = sock.connect_ex((ip_address, port))
if result == 0:
        return True
else:
        return False
except (socket.gaierror, socket.timeout,
ConnectionRefusedError, OSError) as e:
        #Handle common errors
print(f"Port check error: {e}")
return False
except Exception as e:
        print(f"Unexpected port check error: {e}")
return False

if __name__ == "__main__":
    ip_to_check = input("Enter the IP address to check: ")
    if is_ip_alive(ip_to_check):
        print(f"{ip_to_check} is alive.")
        port_to_check = int(input("Enter a port to
check (optional, enter 0 to skip): "))
        if port_to_check > 0:
            if basic_port_check(ip_to_check, port_to_check):
                print(f"Port {port_to_check} is open on {ip_to_check}")
            else:
                print(f"Port {port_to_check} is closed or filtered on {ip_to_check}")
        else:
            print(f"{ip_to_check} is not reachable.")

```

- The python program an input of an ip address from the user and finds the open ports on the provided ip address

Note:

- You need to complete one of the three given Ethical Hacking (EH) projects. Follow these steps while making your report: (ONLY FOR RED TEAMERS)

1. Select and complete ONE project from the given three.
2. Explain your tool's working process and its key functions.
3. Attach relevant screenshots showing your work
4. Upload your project on GitHub and provide the hyperlink in your report.

Learning Outcomes

- **Technical Proficiency:** I gained practical experience in various cybersecurity tools and techniques, including vulnerability scanning, penetration testing, and cryptographic analysis.
- **Understanding of Cybersecurity Lifecycle:** I developed a comprehensive understanding of the steps involved in securing systems, from initial vulnerability assessment to implementing mitigations.
- **Problem-Solving Skills:** Tackling complex security challenges enhanced my analytical thinking and my ability to devise effective solutions under pressure.
- **Professional Development:** My experience improved my ability to work in a team, communicate technical information clearly, and manage time efficiently in a fast-paced environment.

Challenges and Solutions

- **Adapting to Complex Tools:** Initially, mastering advanced cybersecurity tools like Metasploit and Wireshark was challenging. I overcame this by dedicating time to study tutorials and practicing in a simulated environment, which significantly improved my proficiency.
- **Handling Advanced Security Scenarios:** The complexity of real-world security threats required a deep understanding of underlying principles. I tackled this by engaging in continuous learning and seeking guidance from my coordinator and team members.

Conclusion

My internship at Hack Secure was an enriching experience that significantly expanded my knowledge and skills in cybersecurity. The practical exposure to real-world security challenges and the application of advanced tools have solidified my interest in pursuing a career in cybersecurity. This experience has been instrumental in preparing me for the complexities of the cybersecurity field.

Acknowledgments

I express my sincere gratitude to Hack Secure, especially my mentor, Mr. Aman Pandey, and assistant mentor Mr. Prabhat Raj, for their guidance and support throughout my internship. I also thank Amrita Vishwa Vidyapeetham for providing this internship opportunity, which has been crucial in my personal and professional development.

This report encapsulates the essence of my internship experience, highlighting the integration of academic knowledge with practical skills in a professional setting. It reflects my journey of learning, growth, and development in the field of cybersecurity.